

Data Cleaning Report

Apex Manufacturing Executive Dashboard

Project Name: Apex Manufacturing Executive Dashboard

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1. Purpose

The purpose of this document is to describe the data preparation and cleaning activities performed before developing the Power BI dashboard.

Data cleaning ensures that the dataset is accurate, consistent, and suitable for business intelligence reporting by improving data quality and reducing the risk of incorrect analysis.

2. Data Sources

The project uses a structured Microsoft Excel workbook containing manufacturing data from multiple business domains.

The dataset consists of the following tables:

- Plants
 - Products
 - Employees
 - Machines
 - Production
 - Quality
 - Inventory
 - Suppliers
 - Purchase Orders
 - Maintenance
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3. Data Cleaning Objectives

The primary objectives of data cleaning were to:

- Improve data quality.
- Ensure consistency across all tables.

- Standardize data formats.
 - Prepare data for relationship creation.
 - Improve dashboard performance.
 - Support accurate KPI calculations.
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4. Data Cleaning Workflow

The following workflow was followed during data preparation:

Raw Excel Dataset



Data Import into Power BI



Power Query Transformations



Data Validation



Relationship Creation



Data Model



Dashboard Development

5. Data Cleaning Activities

The following validation and transformation activities were performed during the data preparation process.

5.1 Data Type Validation

All columns were reviewed to ensure appropriate data types.

Examples include:

Data Type	Business Example
Whole Number	IDs, Quantities
Decimal Number	Costs, Prices
Date	Production Date, Maintenance Date
Text	Product Name, Plant Name

5.2 Column Standardization

Column names were reviewed and standardized to improve readability and maintain consistency throughout the project.

Examples:

- Plant_ID
- Product_ID
- Machine_ID
- Supplier_ID
- Production_Date

5.3 Relationship Preparation

Primary and foreign key columns were verified before establishing relationships in the Power BI data model.

Relationship validation included:

- Plant_ID
- Product_ID
- Machine_ID
- Supplier_ID

5.4 Date Validation

Date fields were reviewed to ensure they were recognized correctly by Power BI.

This enables:

- Monthly Analysis
 - Quarterly Analysis
 - Yearly Analysis
 - Trend Reporting
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5.5 Numeric Validation

Numeric fields were reviewed to confirm they support calculations used in KPIs and visualizations.

Examples include:

- Production Quantity
 - Maintenance Cost
 - Procurement Cost
 - Unit Cost
 - Selling Price
 - Available Stock
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5.6 Text Validation

Text fields were reviewed to ensure consistency and avoid issues caused by formatting differences.

Examples include:

- Plant Names
 - Product Categories
 - Machine Types
 - Departments
 - Supplier Names
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6. Data Quality Checks

The following quality checks were completed prior to dashboard development.

Validation	Status
Table Structure Reviewed	Completed
Column Names Standardized	Completed
Data Types Validated	Completed
Primary Keys Reviewed	Completed
Foreign Keys Reviewed	Completed
Date Fields Validated	Completed
Numeric Fields Reviewed	Completed
Relationship Readiness Verified	Completed

7. Data Integrity

The cleaned dataset maintains referential integrity across all business entities.

Relationships were validated between:

- Plants and Production
- Products and Production
- Products and Inventory
- Suppliers and Purchase Orders
- Machines and Maintenance
- Products and Quality

This ensures accurate filtering and aggregation within Power BI.

8. Benefits of Data Cleaning

The data preparation process provides several advantages:

- Improved reporting accuracy
 - Better dashboard performance
 - Consistent business calculations
 - Reliable KPI measurements
 - Reduced data inconsistencies
 - Simplified maintenance of the data model
 - Enhanced user confidence in analytical results
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9. Best Practices Followed

The following best practices were applied during data preparation:

- Standardized naming conventions
 - Consistent data types
 - Verified relationships
 - Business-friendly column names
 - Structured fact and dimension tables
 - Clean and maintainable data model
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10. Output

After completing the data cleaning process, the dataset was ready for:

- Data Modeling
- Relationship Creation

- DAX Measure Development
 - KPI Calculation
 - Interactive Dashboard Design
 - Executive Reporting
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11. Conclusion

The data cleaning process established a consistent and reliable foundation for the Apex Manufacturing Executive Dashboard. By validating data types, preparing relationships, standardizing business entities, and ensuring analytical readiness, the dataset became suitable for enterprise-level reporting in Microsoft Power BI.

A structured data preparation process improves dashboard performance, increases reporting accuracy, and ensures that business decisions are based on trustworthy information.